HISTOPATHOLOGY

The histopathology of Kaposi sarcoma (KS) varies according to the stage of disease development.

Patch Stage:

Early patch-like lesions show subtle histopathologic changes, primarily characterized by an increase in the number of dermal vessels lined by slightly atypical endothelial cells (Fig. 128-5). These vessels are predominantly located in the superficial dermis, run parallel to the skin surface, and often form irregular, slit-like or cleft-like spaces. Focal hemosiderin deposits, extravasated red blood cells, and a moderate inflammatory infiltrate composed of lymphocytes, histiocytes, and plasma cells are commonly seen in the surrounding tissue. Key differential diagnoses at this stage include **lymphangioma** and **granulation tissue**.

Plaque Stage:

KS plaques exhibit more distinctive features, with extensive vascular proliferation throughout all levels of the dermis. There are numerous dilated and angulated vascular spaces that dissect collagen bundles, creating a spongy network of residual collagen. A hallmark of papular and plaque lesions is the presence of solid cords and fascicles of spindle cells intermixed with jagged vascular channels. This biphasic pattern—comprising both angiomatous and solid components—gradually evolves into a predominantly sarcomatous morphology as the disease progresses.

Nodular Stage:

Nodular lesions are largely composed of proliferating spindle cells arranged in interlacing bundles and fascicles, with scattered, irregular, slit-like vascular spaces lacking well-defined endothelial linings. With advancing disease, these lesions may display marked nuclear pleomorphism, atypia, and increased mitotic activity. At the periphery of such tumors, areas resembling earlier KS stages—featuring abnormal vascular lumina, intravascular and extravasated erythrocytes, and siderophages—may persist. Erythrocytes are often entrapped within clefts formed by spindle cells and appear as eosinophilic globules; erythrophagocytosis can occasionally be observed. A mixed inflammatory infiltrate (lymphocytes, histiocytes, plasma cells, and occasionally neutrophils) is consistently present across all stages.

FIGURE 128-5: Biopsy from a nodular Kaposi sarcoma lesion showing both angiomatous areas resembling lymphangioma and solid, sarcomatous fascicles.

DIFFERENTIAL DIAGNOSIS

Differential diagnoses for cutaneous and mucosal Kaposi sarcoma vary by clinical stage:

Most Likely

- **Localized early disease:**
- Hematoma
- Stasis dermatitis
- Bacillary angiomatosis
- Pyogenic granuloma
- Well-differentiated angiosarcoma
- **Advanced disease:**
- Acroangiodermatitis (pseudo-Kaposi)
- Bacillary angiomatosis
- **Oropharyngeal lesions:**
- Non-Hodgkin lymphoma
- Squamous cell carcinoma
- Bacillary angiomatosis

Consider

- **Localized disease:**
- Dermatofibroma
- Melanoma
- Dermatofibrosarcoma protuberans
- Leiomyosarcoma
- Kaposiform hemangioendothelioma (in children)
- Blue rubber bleb nevus syndrome

- **Advanced disease:**

- Metastatic melanoma
- Cutaneous metastases from internal malignancies or leukemias
- Erythema elevatum et diutinum
- Angiosarcoma

Always Rule Out

- HIV-1 infection
- Other causes of immunosuppression
- Melanoma

TREATMENT

Treatment strategies for Kaposi sarcoma depend on the clinical type, extent of lesions, and organ systems involved.

Despite strong evidence implicating **human herpesvirus-8 (HHV-8)** as the causative oncogenic agent in KS, antiherpetic therapies have not demonstrated consistent clinical efficacy and remain largely experimental.